

Julian Karlbauer M.Sc.

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Applied Researcher & Engineer. My research interests lie at the intersection of spatial computing, digital fabrication, and intelligent interactive systems, with a focus on empowering and augmenting human abilities using an applied, systems-focused, and interdisciplinary approach. Expert in XR Software Engineering, and building research prototypes for industrial scale, with experience at Siemens, Audi, and Silicon Valley startups.

EXPERIENCE

Sabbatical

Latin America, Middle East, Europe, Asia (06/2025 – present)

Planned sabbatical focused on gaining perspective through cross-cultural engagement, journalism, independent projects, and clarifying my long-term vision as a researcher.

Phygtl Inc. Founding Engineer

Palo Alto, USA (remote) (12/2024 – 06/2025)

Equity based Founding Role in an XR startup dedicated to bridging physical and digital spaces through collaborative social AR experiences and enabling Generative-AI based 3D co-creation through real-world visual input. My responsibilities as a Founding Engineer included strategic decision-making in the design of software architecture, direct collaboration with the executive team (CEO, CTO) to define the technical roadmap, leading the operational Quality Assurance, and contributing significantly to the code base.

Ulm University Research Associate

Ulm, Germany (01/2024 – 10/2024)

Worked as a Research Associate under Prof. Dr. Enrico Rukzio on *Personal Fabrication*. My research focus was on Generative-AI-supported 3D authoring in AR to enhance Digital Manufacturing workflows. As part of my work, I attended ACM SCF '24 and demoed *MixGen*, a working AR prototype (Python, C#) for generative 3D content authoring from multi-modal real-world input. I further attended ACM SCF Summer School and co-published *VRCreatIn*, a tablet-based 3D content authoring VR tool, at ACM MUM '24 [3].

Siemens AG Master's Thesis Student

Munich, Germany (04/2023 – 11/2023)

I conducted my Master's Thesis on enhancing Quality Assurance in Additive Manufacturing processes through data-driven Digital Twin and Augmented Reality Interfaces. I designed, built, and evaluated an AR interface (C#, Python, JavaScript) for an industrial metal 3D printer (WAAM), enabling interactive and in-situ visualisation of Digital Twin data, such as CAD and spatio-temporal anomaly prediction. The prototype received high institutional validation at Siemens and was recognised as a future candidate for demonstration to the CEO and at the Formnext exhibition.

Audi AG Software Engineering Intern

Ingolstadt, Germany (03/2022 – 09/2022)

Worked in the HCI team of the Audi Production Lab to develop a mobile cross-device Augmented Reality platform designed for Audi internal communication on 3D Digital Twin data. I was responsible for iOS-facing Unity3D development, Interaction Design, and assistance in conducting interviews and user-studies.

TriCAT GmbH R&D Intern

Ulm, Germany (03/2021 – 06/2021)

Proposed and built a machine vision research prototype for in-situ real-time object recognition and 3D localisation on HoloLens 2 using manually fine-tuned CNNs (Python, Tensorflow) and environment meshes with a compute-efficient remote GPU architecture, pioneered applying Deep Learning tools for problem-solving in the company.

Ulm University Research Assistant

Ulm, Germany (04/2018 – 03/2021)

Significantly contributed to rigorous research published in ACM CHI and ACM ToCHI, with a focus on 3D authoring [1] and health aspects in Virtual Reality [2]. My responsibilities included XR prototyping, contributing to design processes, providing novel ideas to research projects, conducting user studies, creating footage and media for publications.

EDUCATION

Ulm University *M. Sc. in Media Informatics*

Ulm, Germany (10/2020 – 03/2024)

Thesis Title: *“Show, Don’t Tell: Enhancing Quality Assurance in Additive Manufacturing through Augmented Reality”* (Thesis Grade 1.0; Overall Grade: 1.2)

NTNU *Visiting Student in Generative AI*

Trondheim, Norway (08/2021 – 12/2021)

Thesis Title: *“Raw Audio Music Emotion Composition using Mood Annotated Spotify Playlists”*
(Thesis Grade 1.0; Overall Grade: 1.0)

Ulm University *B. Sc. in Media Informatics*

Ulm, Germany (04/2016 – 07/2020)

Thesis Title: *“An Exploration of Foveated Blue Light Filters in Virtual Reality”*
(Thesis Grade 1.0)

ACADEMIC SERVICES

- Conference Reviewer CHI 2025
- Served as a student volunteer at the ACM SCF 2024 conference.

ACADEMIC PUBLICATIONS

- [1] Tobias Drey, Nico Rixen, **Julian Karlbauer**, and Enrico Rukzio. 2024. VRCreatIn: Taking In-Situ Pen and Tablet Interaction Beyond Ideation to 3D Modeling Lighting and Texturing. In Proceedings of the International **Conference on Mobile and Ubiquitous Multimedia (MUM '24)**. Association for Computing Machinery, New York, NY, USA, 24–35. <https://doi.org/10.1145/3701571.3701580>
- [2] Teresa Hirzle, Fabian Fischbach, **Julian Karlbauer**, Pascal Jansen, Jan Gugenheimer, Enrico Rukzio, and Andreas Bulling. 2022. Understanding, Addressing, and Analysing Digital Eye Strain in Virtual Reality Head-Mounted Displays. **ACM Trans. Comput.-Hum. Interact.** 29, 4, Article 33 (August 2022), 80 pages. <https://doi.org/10.1145/3492802>
- [3] Tobias Drey, Jan Gugenheimer, **Julian Karlbauer**, Maximilian Milo, and Enrico Rukzio. 2020. VRSketchIn: Exploring the Design Space of Pen and Tablet Interaction for 3D Sketching in Virtual Reality. In Proceedings of the 2020 **CHI Conference on Human Factors in Computing Systems (CHI '20)**. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376628>